Confidence interval for the difference of two means. Independent samples, variance known

University example

Background You have the dataset from the lesson.
Task 1 Calculate the 99% confidence interval
Task 2 Compare it to the 95% confidence interval from the lesson

	Engineering	Management	Difference
Size	100	70	?
Sample mean	58	65	-7.00
Population std	10	5	1.16

99% CI@z=0.005	0.995	Look up in z-table
0.005	2.58	

Confidence Interval

Z	CI low	CI high
99%	-10.01	-3.99

Confidence intervals

Z	CI low	CI high
95%	-9.28	-4.72

A higher confidence leads to a broader interval